



AEGIS

Anonymization & Exchange Gateway
for Imaging Studies

Brain MRI

PET

CT

DICOM

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A GCP-native platform for secure, HIPAA-compliant sharing of medical imaging data across all DICOM modalities

In Greek mythology, the aegis was the divine shield of Zeus and Athena — a symbol of protection. The name captures our mission: shielding patient identity while enabling the free flow of imaging data for research.

The Problem

Hospitals and research institutions need to share medical imaging data — MRI, CT, PET, ultrasound, X-ray, mammography, nuclear medicine, and more — for multi-site clinical trials, research collaborations, and second opinions. Today this requires:

- **Installing site software** — platforms like XNAT require a Java-based desktop client or server daemon installed and IT-approved at every participating hospital. MIRC CTP requires a standalone Java application. Even simpler tools need command-line setup by hospital technical staff.
- **Manual de-identification** prone to human error, leaving PHI exposed. A 2024 NCI benchmark challenge (MIDI-B) confirmed that inconsistencies in DICOM metadata and burned-in PHI remain unsolved problems even for specialized tools.
- **No automated defacing** — facial features in 3D head scans (MRI, CT) enable re-identification via facial reconstruction software, yet most platforms offer no defacing pipeline.
- **Self-hosted infrastructure burden** — XNAT and Orthanc require dedicated servers, database maintenance, and ongoing IT support at the receiving institution.
- **Fragmented compliance** — each site manages its own HIPAA controls, audit trails, and BAAs, with no centralized visibility.
- **Modality limitations** — most platforms were built for a single specialty (XNAT for neuroimaging, commercial tools for cardiology) and don't generalize cleanly to other DICOM data.

Result: Data sharing is slow, risky, and expensive. Multi-site studies are delayed by months waiting for IT approvals and software installations at each new site.

Market Opportunity

Medical imaging data sharing sits at the intersection of two fast-growing markets:

Market	2024 Size	Projected	CAGR
Medical Image Exchange Systems	\$4.3B	\$8.7B (2034)	8.1%
Enterprise Imaging IT	—	—	12.2% (2025–2030)

Medical Digital Imaging Systems (total)	\$36.1B	\$38.3B (2025)	—
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Why now:

- NIH, NSF, and FDA are all mandating data sharing for publicly funded research — but tooling lags far behind policy.
- A 2024 NCI/MICCAI challenge (MIDI-B) formally benchmarked de-identification tools against HIPAA Safe Harbor + DICOM Confidentiality Profiles — confirming that no existing open-source tool handles all cases reliably.
- Cloud-native healthcare platforms (GCP Healthcare API, AWS HealthImaging) have matured to the point where managed DICOM storage is now commodity infrastructure — reducing build risk.
- Growing multi-site clinical trial volume (particularly in oncology and neurology) is driving demand for scalable, zero-friction data collection from distributed hospital networks.

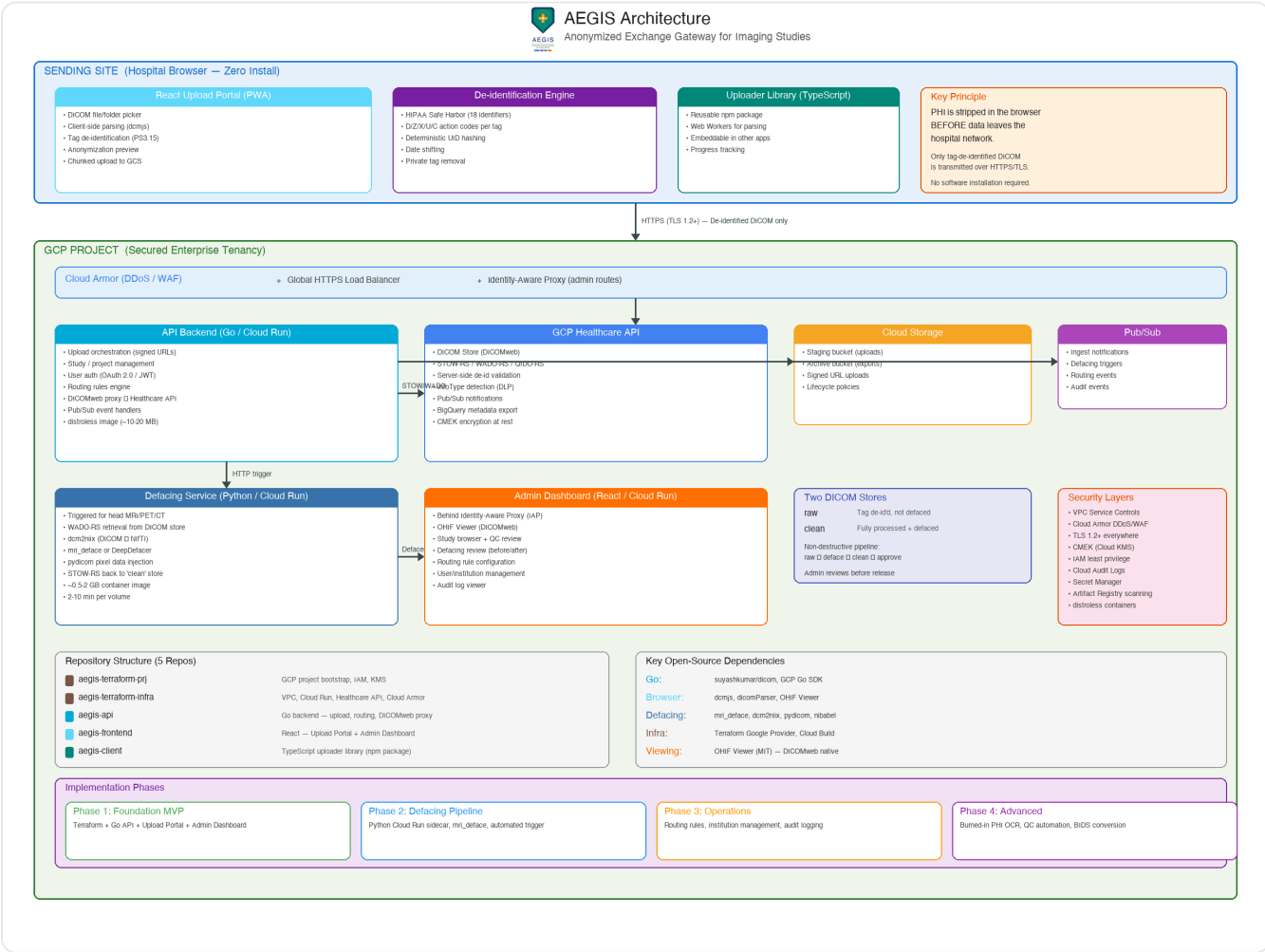
The Solution

AEGIS is a cloud-hosted platform that makes secure medical image sharing as easy as uploading a file — for **any DICOM modality**.

Capability	How It Works
Zero-install anonymization	Browser-based DICOM tag de-identification. No software to install at hospital sites. Works with any DICOM data.
Modality-agnostic	Supports all DICOM modalities: MRI, CT, PET, PET/CT, US, X-ray, mammography, NM, SPECT, RT, and more.
Automated defacing	Server-side facial feature removal from head scans. Non-head studies bypass defacing automatically.
HIPAA by design	Client-side PHI stripping before data leaves the hospital network. VPC-SC, CMEK, audit trails.
Managed DICOM storage	GCP Healthcare API — DICOMweb endpoints, no servers to maintain.

AI-powered QC	Vertex AI detects burned-in PHI, assesses image quality, flags issues automatically.
Admin review	OHIF web viewer for QC, defacing review, and routing — all in the browser.

Architecture



How It Works

1. Uploader opens AEGIS in their browser (no install)
2. Drags DICOM files into the upload portal (any modality)

3. Browser parses and strips all PHI tags (PS3.15 Annex E)
4. Uploader reviews before/after anonymization preview
5. Confirms → encrypted upload to GCP (de-identified only)
6. Server validates de-identification completeness
7. Head scans → auto-defacing; others → clean store
8. Admin reviews in OHIF viewer, approves or rejects
9. Approved study routed to destination institution

Key: Patient-identifying data **never leaves the hospital network**. The de-identification engine follows [DICOM PS3.15 Annex E](#) — the official industry standard that defines exactly which data elements to remove, zero out, or replace to strip all 18 HIPAA Safe Harbor identifier categories from medical images. The process is **modality-agnostic** — it works identically whether the study is a brain MRI, a chest X-ray, or a cardiac ultrasound.

Supported Modalities

Category	Modalities
Cross-sectional	MRI, CT, PET, PET/CT, PET/MR
Projection	X-Ray (CR/DR), Mammography (FFDM/DBT), Fluoroscopy
Nuclear Medicine	SPECT, Planar Scintigraphy
Ultrasound	Echocardiography, Abdominal, Vascular, OB/GYN
Radiation Therapy	RT Structure, RT Plan, RT Dose, RT Image
Other DICOM	Secondary Capture, Structured Reports, Key Images, Presentation States

MVP focus: Brain MRI, PET, and CT — these exercise the full pipeline including automated defacing. All other modalities use the same upload and de-identification flow.

Technology Stack

Layer	Technology	Why
Backend	Go on Cloud Run	~10 MB container, 100ms cold starts, minimal CVE surface
Frontend	React + TypeScript	Modern, fast, PWA-capable
Database	Cloud SQL (PostgreSQL 15)	Relational data: users, projects, audit trail
DICOM Storage	GCP Healthcare API	Managed DICOMweb, built-in de-id validation
Analytics	BigQuery	DICOM metadata export, compliance reporting
AI/ML	Vertex AI	Burned-in PHI detection, image QC automation
Defacing	Python (mri_deface)	Proven facial feature removal, isolated sidecar
Viewer	OHIF Viewer	MIT-licensed, DICOMweb-native, all modalities
Infrastructure	Terraform	Reproducible, auditable IaC
Email	PSC→on-prem SMTP (internal) + SendGrid (external)	Dual-path, no PHI in notifications

Competitive Landscape

	AEGIS	XNAT	Flywheel	MIRC CTP
Hosting	GCP managed	Self-hosted	Commercial SaaS	On-prem
Client install	None (browser)	Desktop app	CLI tool	Java app

Modalities	All DICOM	Neuroimaging	All imaging	All imaging
Defacing	Automated	Manual/plugin	Built-in	None
Open source	Yes	Yes	No	Yes
DICOM store	Healthcare API	PostgreSQL + FS	MongoDB + S3	Filesystem

Differentiators:

1. Zero-install browser anonymization for any DICOM modality
2. Modality-agnostic — not locked to a single specialty
3. GCP-native with Healthcare API — managed DICOMweb, built-in de-id, BigQuery analytics
4. Automated defacing pipeline for head imaging
5. Vertex AI for burned-in PHI detection and image QC
6. Minimal attack surface (Go + distroless)

Phased Roadmap

Phase 1 — MVP (Brain MRI/PET/CT)

- GCP project + infrastructure (Terraform)
- Upload Portal with browser-based DICOM anonymization (all modalities)
- Go API with signed upload and Healthcare API ingest
- Admin Dashboard with OHIF viewer for QC
- Cloud SQL for app data, BigQuery for analytics

Phase 2 — Defacing + Notifications

- Automated mri_deface pipeline on Cloud Run (head imaging)
- Before/after comparison in admin dashboard
- Email notifications: SendGrid (external), SMTP/PSC (internal)

Phase 3 — Operations + Multi-Modality Expansion

- Multi-project routing engine for all modalities
- Institution onboarding and RBAC
- Configurable anonymization profiles per modality/project
- Audit reporting and compliance dashboards

Phase 4 — AI/ML

- Vertex AI burned-in PHI detection (all modalities)
- Automated image quality assessment
- Smart routing with modality/anatomy classification
- BIDS format conversion for research output

Cost Estimate (Dev/POC)

Resource	Monthly Cost
Cloud Run (API + dashboard)	~\$5-15 (free tier covers dev)
Cloud SQL (db-f1-micro)	~\$10
Healthcare API (DICOM store)	~\$0.50/GB stored
Cloud Storage (staging)	~\$0.02/GB
BigQuery	Free tier (1 TB queries/month)
Vertex AI	Pay per prediction (minimal during dev)
SendGrid	Free tier (100 emails/day)
Total (dev)	~\$20-40/month

Production costs scale with data volume. A typical multi-site research study with 1,000 imaging sessions (~500 GB) would cost approximately \$50-100/month in storage.



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Ready to Build

AEGIS enables secure, zero-install medical image sharing across all DICOM modalities — with automated anonymization, intelligent defacing, and AI-powered quality control — built on GCP with HIPAA compliance from day one.